

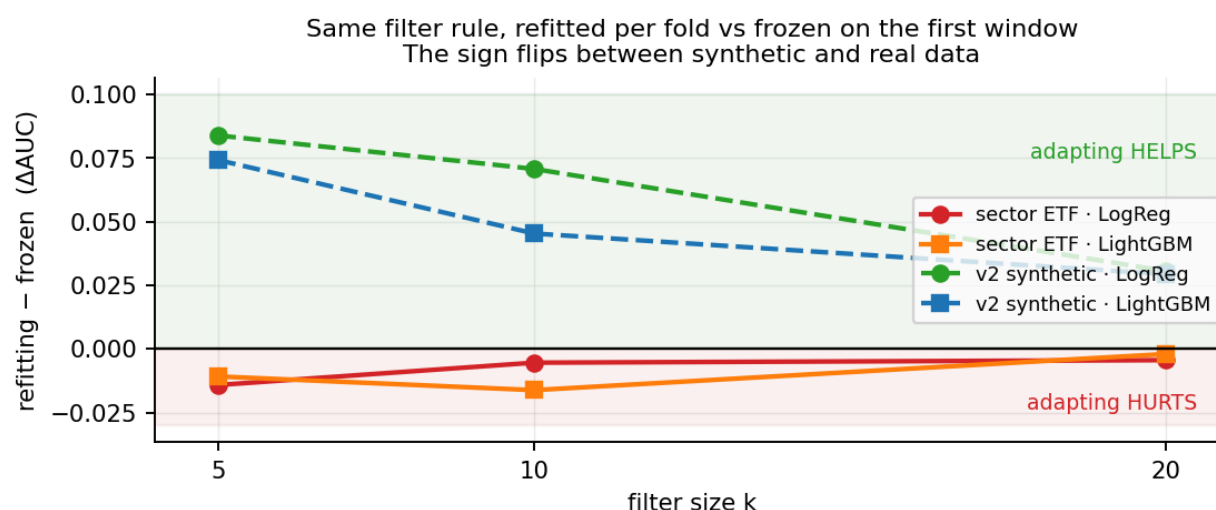
# Phase 3 & 4 — Supporting Results

Supplement to the update email. Every number was produced by running the current codebase; nothing is quoted from an earlier generation.

**Sample sizes, stated up front.** Phase 3 uses 5 seeds; at  $n=5$  the minimum attainable two-sided Wilcoxon  $p$  is 0.0625, so the significance tests below are one-sample t-tests against a deterministic comparator. Phase 4 is one run per cell — the filter and baseline arms are deterministic, so they have no seed variance to report. Two datasets throughout. These are strong directional results, not a 30-seed protocol.

## 1. Adaptation hurts on real data

Same filter rule, same  $k$ , one refitted on every fold's training window and one frozen on the first window and never updated.



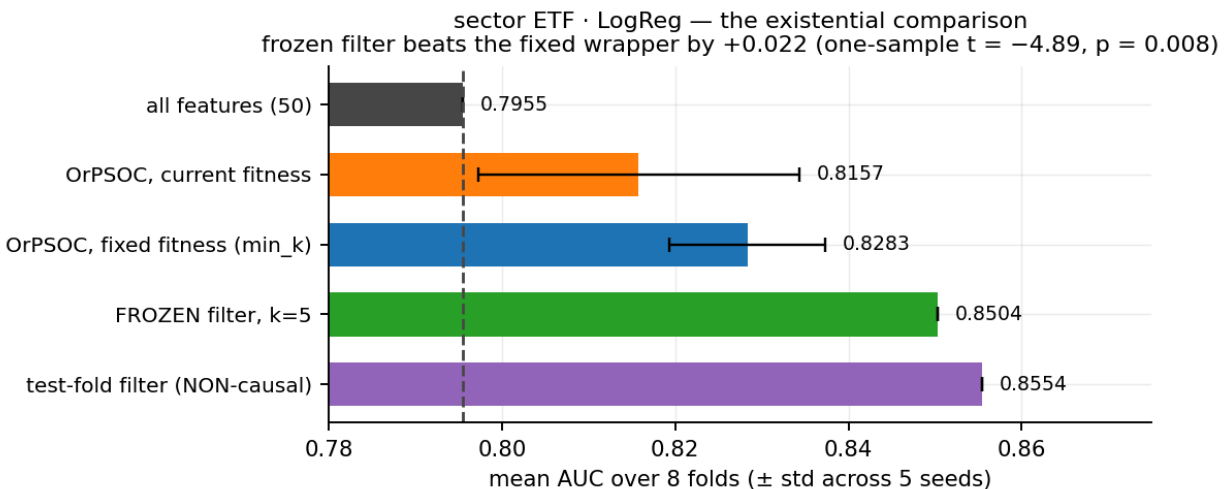
|                         | k=5     | k=10    | k=20    | frozen wins |
|-------------------------|---------|---------|---------|-------------|
| sector ETF · LogReg     | −0.0141 | −0.0054 | −0.0044 | 3/3         |
| sector ETF · LightGBM   | −0.0108 | −0.0162 | −0.0020 | 3/3         |
| v2 synthetic · LogReg   | +0.0839 | +0.0708 | +0.0306 | 0/3         |
| v2 synthetic · LightGBM | +0.0742 | +0.0454 | +0.0294 | 0/3         |

The sign flips cleanly, 6/6 real cells against 6/6 synthetic cells.

**Mechanism.** The synthetic benchmark *defines* regime shift as the informative feature set rotating: `s0–s2` are predictive only before the switch and `s3–s4` only after (measured on v2 — pre-switch  $\text{AUC}(s0-s2)=0.938$  vs  $\text{AUC}(s3-s4)=0.514$ ; post-switch  $0.545$  vs  $0.943$ ). Adaptation is therefore rewarded by construction. Real financial regime shift moves the distribution without changing *which* features are informative — sector volatility and momentum stay informative through 2008 and 2020 — so re-ranking each fold chases sampling noise in the most recent window and discards a stable, correct answer.

**Caveat for framing.** One real dataset supports this. Before it headlines the paper it should be reproduced on Fama-French and ideally one further market.

## 2. Phase 3 — the existential comparison



sector ETF, LogReg final model, mean AUC over 8 folds,  $\pm$  std across 5 seeds:

| arm                             | AUC                                   | vs baseline    | n_sel    |
|---------------------------------|---------------------------------------|----------------|----------|
| all features                    | 0.7955 $\pm$ 0.0000                   | —              | 50       |
| OrPSOC, current fitness         | 0.8157 $\pm$ 0.0185                   | +0.0202        | 13.0     |
| OrPSOC, fixed fitness ( min_k ) | 0.8283 $\pm$ 0.0090                   | +0.0328        | 14.2     |
| <b>FROZEN filter, k=5</b>       | <b>0.8504 <math>\pm</math> 0.0000</b> | <b>+0.0549</b> | <b>5</b> |
| test-fold filter (NON-causal)   | 0.8554 $\pm$ 0.0000                   | +0.0599        | 10       |

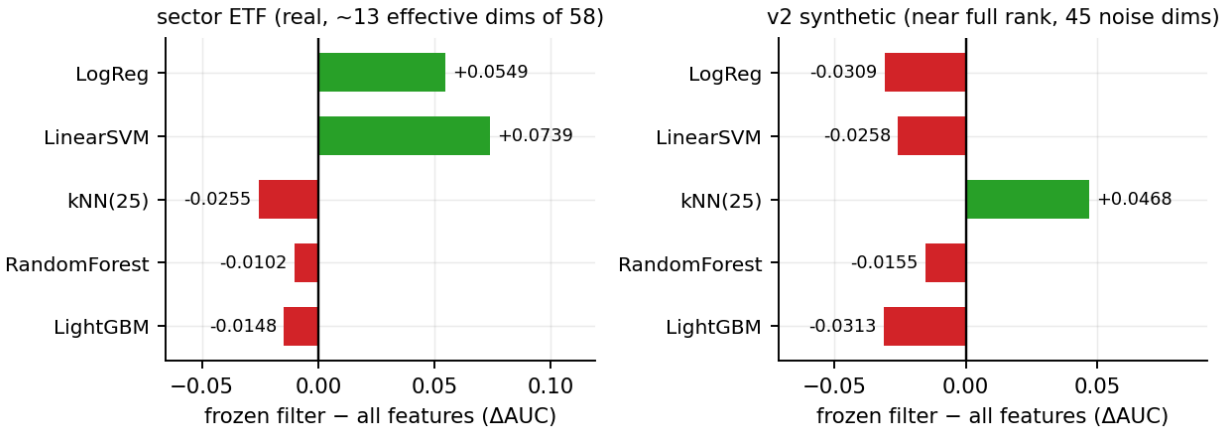
- frozen filter vs OrPSOC( min\_k ): **+0.0220**, one-sample  $t = -4.89$ , **p = 0.008**
- frozen filter vs OrPSOC(current): **+0.0346**,  $t = -3.73$ , **p = 0.020**
- fixing the compass recovered **36%** of the gap (+0.0126, paired  $p = 0.29$ )

The frozen filter is deterministic — no seed variance — and lands within 0.005 of a filter that was allowed to see the test fold.

**Note on a corrected label.** Earlier experiments in this project called the frozen filter an “oracle”. It is not: it is causal, uses only the first training window, and is fully achievable. That mislabelling understated the threat, by presenting an achievable method as an unreachable bound. The genuinely non-causal arm is the last row, and even that is only an upper bound on *univariate filtering*, not on selection in general.

### 3. Phase 4 — collinearity, not capacity

What does selection buy each classifier? Linear models gain; trees and kNN lose



Frozen filter minus all-features, per classifier:

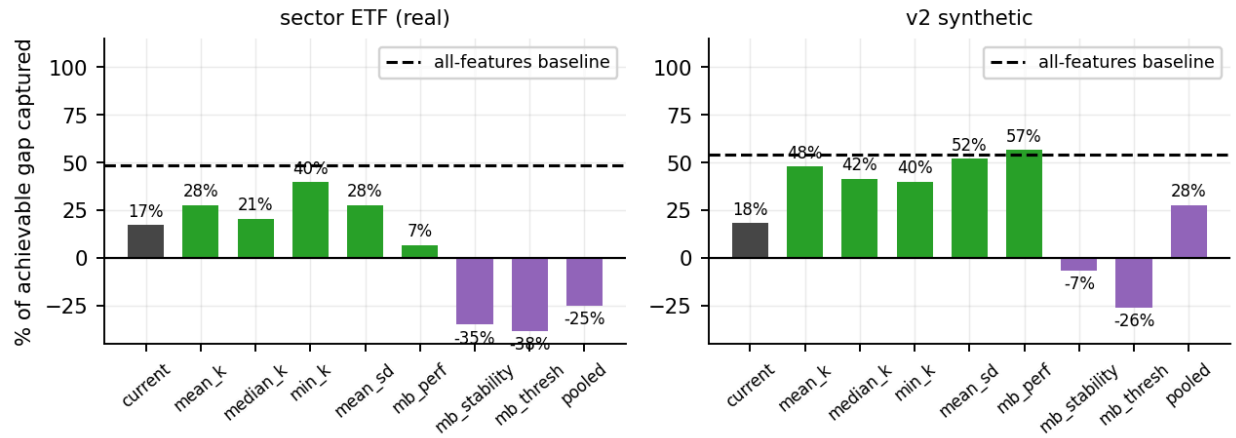
| classifier   | sector ETF (real) | v2 synthetic   |
|--------------|-------------------|----------------|
| LogReg       | <b>+0.0549</b>    | -0.0309        |
| LinearSVM    | <b>+0.0739</b>    | -0.0258        |
| kNN(25)      | -0.0255           | <b>+0.0468</b> |
| RandomForest | -0.0102           | -0.0155        |
| LightGBM     | -0.0148           | -0.0313        |

kNN has no built-in feature selection at all, so the capacity hypothesis predicts a large gain — and it **loses** 0.026 on real data. The variable that does sort the table is **collinearity sensitivity**: sector ETF has 58 nominal features spanning ~13 effective dimensions, which inflates coefficient variance in linear models but leaves trees (splitting) and kNN (distance aggregation) unaffected. Pruning to 5 features stabilises the linear fit and throws information away for the others.

v2 shows the mirror image because it has almost no collinearity but 45 pure-noise dimensions, which wreck a distance metric — hence kNN is the only gainer there.

## 4. The criterion bake-off

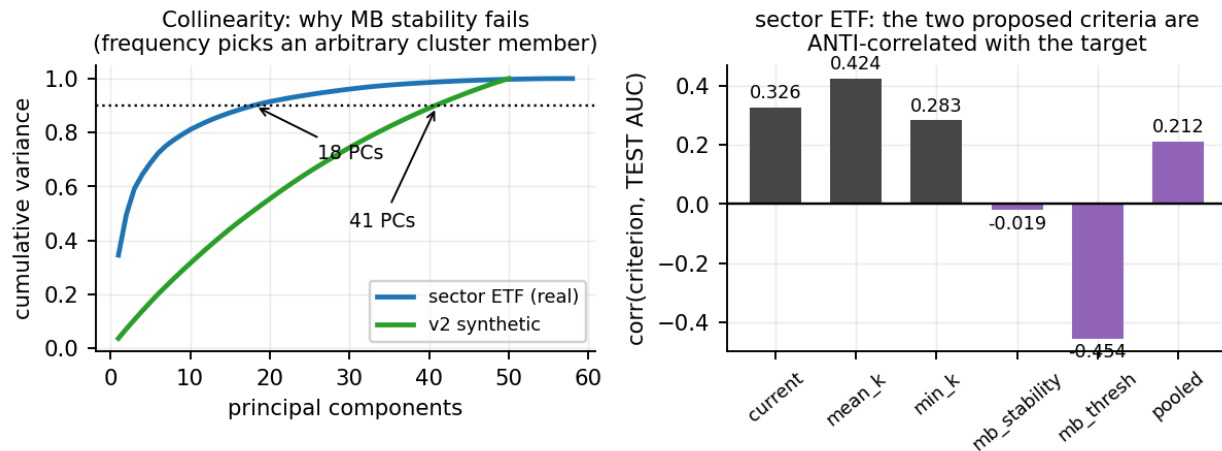
Fitness-criterion bake-off (purple = the two newly proposed criteria)



Criterion isolated from search: 100 candidate subsets per fold, scored causally, then “what would a *perfect* search maximising this land on?” Ranked by percentage of the achievable gap captured (0% = no better than random picking).

| criterion                    | sector ETF  | v2 synthetic |
|------------------------------|-------------|--------------|
| min_k                        | <b>40%</b>  | 40%          |
| mean_k                       | 28%         | 48%          |
| mean_sd                      | 28%         | <b>52%</b>   |
| median_k                     | 21%         | 42%          |
| current                      | 17%         | 18%          |
| mb_perf                      | 7%          | <b>57%</b>   |
| <b>pooled</b>                | <b>-25%</b> | 28%          |
| <b>mb_stability</b>          | <b>-35%</b> | -7%          |
| <b>mb_thresh</b>             | <b>-38%</b> | -26%         |
| <i>all-features baseline</i> | <i>49%</i>  | <i>54%</i>   |

Both newly proposed criteria are the worst on real data, and negative percentages mean they select subsets *worse than picking at random*.



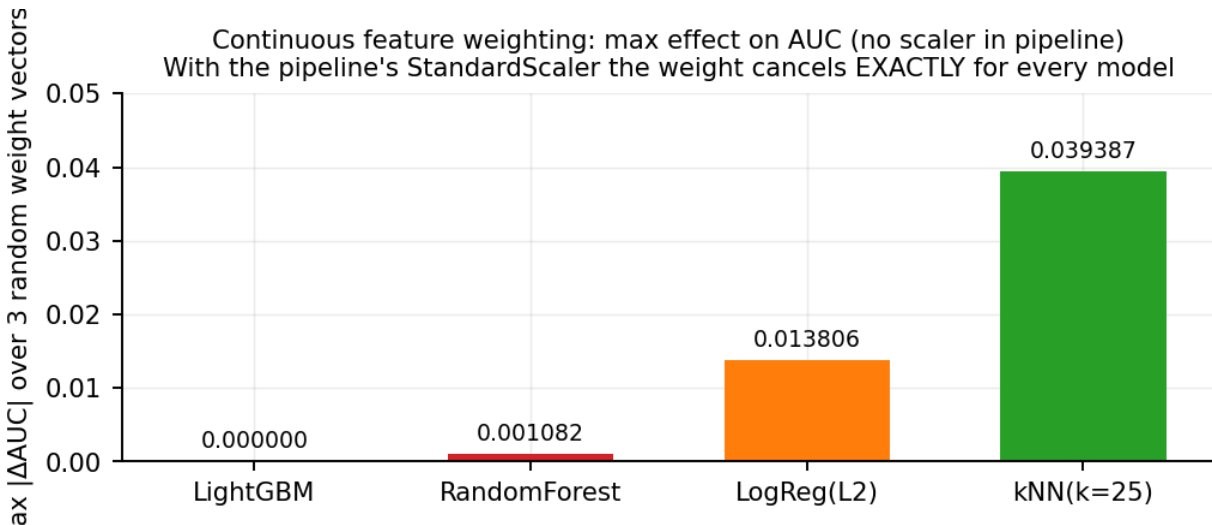
**Why MB stability fails.** Correlation with test AUC is  $-0.019$  (mean form) and  $-0.454$  (classical threshold form) — actively anti-correlated. With 58 features in  $\sim 13$  effective dimensions, selection frequency measures which member of a collinear cluster arbitrarily won a bootstrap resample, not which features are useful; thresholding then concentrates on the in-sample-strongest, least diverse set. This is not an implementation artefact: the first version scored by *mean* frequency and was size-blind (its argmax picked  $k \approx 5.5$  against  $\sim 16$  for every size-neutral criterion); rebuilding it in the classical threshold form made the result **worse**, which rules that explanation out.

**Why pooled validation fails.** On sector ETF the HMM regime state is **0.392 correlated with the target** (y-mean 0.32 vs 0.738 across states) — the target is a rolling-volatility threshold and the regime is a rolling-volatility state, so conditioning on regime strips out the label variation. On v2, where that correlation is 0.010, pooled works (28%). The idea is sound; it is incompatible with *this dataset's* target definition.

**No criterion wins on both datasets, and none reaches the baseline on real data.**

`min_k` is the most consistent (40% on both) but it then hurt in Phase 3 on three of four cells:  $+0.0126$  ( $p=0.29$ ) on sector ETF·LogReg,  $-0.0152$  on sector ETF·LightGBM,  $-0.0620$  ( $p<0.001$ ) on v2.

## 5. Continuous feature weighting is mathematically inert



Max |ΔAUC| over three random weight vectors in [0.05, 1], **with no scaler in the pipeline**:

| model        | max change |
|--------------|------------|
| LightGBM     | 0.000000   |
| RandomForest | 0.001082   |
| LogReg (L2)  | 0.013806   |
| kNN (k=25)   | 0.039387   |

For LightGBM the change is exactly zero: trees split on thresholds, and scaling a column is a monotone transform, so no reachable split point changes.

The stronger result is that the pipeline standardises:

StandardScaler(X) vs StandardScaler(X · w): max abs difference = 2.2e-15

$$z = (w \cdot x - \text{mean}(w \cdot x)) / \text{std}(w \cdot x) = w(x - \text{mean } x) / (w \cdot \text{std } x) = (x - \text{mean } x) / \text{std } x$$

**The weight cancels exactly, for every model.** A continuous-weight OrPSOC on this pipeline would search a space in which the objective is literally constant.

This does not rule out the *hyperparameter* variant, which changes the model in ways weighting cannot and tunes the embedded selector rather than competing with it. That direction inherits the compass problem, since per-regime tuning relies on the same trailing-window validation that §1 shows is counterproductive.

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## 6. What is not yet done

- The full L1–L4 ablation has **not** been re-run on the v2 benchmark.
- The `min_k` default is **not** locked, pending the ruling in the email.
- The LogReg Paradox narrative is **unwritten**, pending the same ruling.
- Three pending changes each alter reported numbers and should go into one regeneration run: v2 data, `fold_recall_active`, and  $\theta = 0.5$ .

### ON `FOLD_RECALL_ACTIVE`

A metric defect was found and fixed while auditing L4. Union recall over all five signal features can reach 1.0 only by keeping the two or three features that have gone inert after the switch; a selector that correctly drops them scores at most  $3/5 = 0.60$  pre-switch and  $2/5 = 0.40$  post-switch. **The union denominator pays for dead weight and penalises correct adaptation** — the exact behaviour the study exists to measure.

`fold_recall_active` scores against the features actually predictive in each fold's regime, so the attainable maximum is 1.0 everywhere; straddling folds return NaN rather than a number that would silently mean two different things. Since §3.6 leans on recall to make the Jaccard result conclusive, that section needs regenerating and the direction of the effect could change.